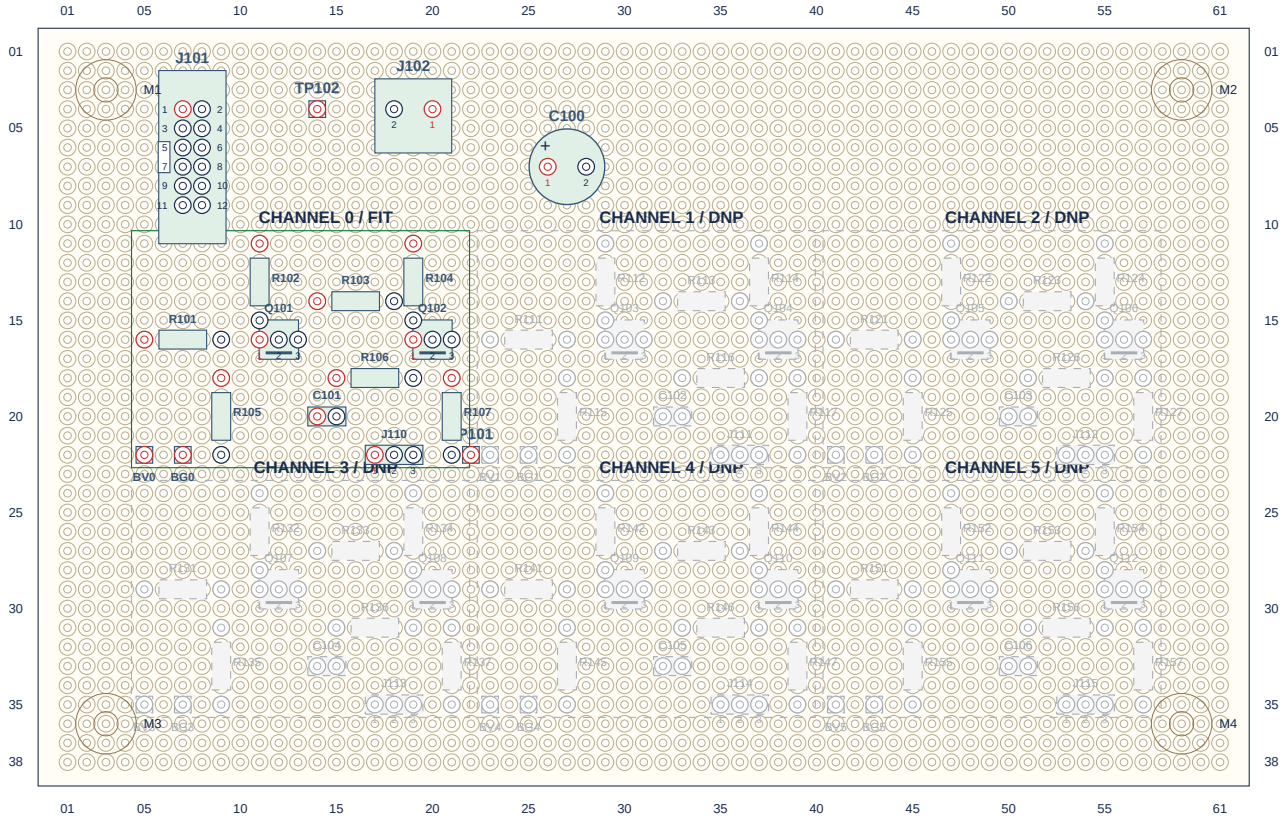
Servo-output module: component placement

Component side; copper underneath. Green = first assembly. Grey = reserved/DNP. Hole IDs: C(column)-R(row).



FIRST ASSEMBLY

J101
IDC input: odd pins S0..S5
J102
Servo power: 1 positive / 2 GND
C100
470uF/35V: D10 x H20, 5.00-mm leads
Q101 / Q102
BC547B: 1 C / 2 B / 3 E
R101 / R103
4.7k
R102 / R104
1k
R105 / R107
100k
R106
180 ohm
C101
100nF
J110
Servo0: 1 signal / 2 positive / 3 GND
TP101 / TP102
Servo0 signal / GND
BV0 / BG0
Existing-pad power-wire junctions

C100 leads gently formed to 5.08 mm.

J102: widen C20-R04/C18-R04 to 1.3 mm.

Channels 1..5: do NOT populate yet.

Full six-servo power sizing still pending.

Only 32 wires are needed for the first assembly. Fit common parts and channel0; keep DNP pads free.

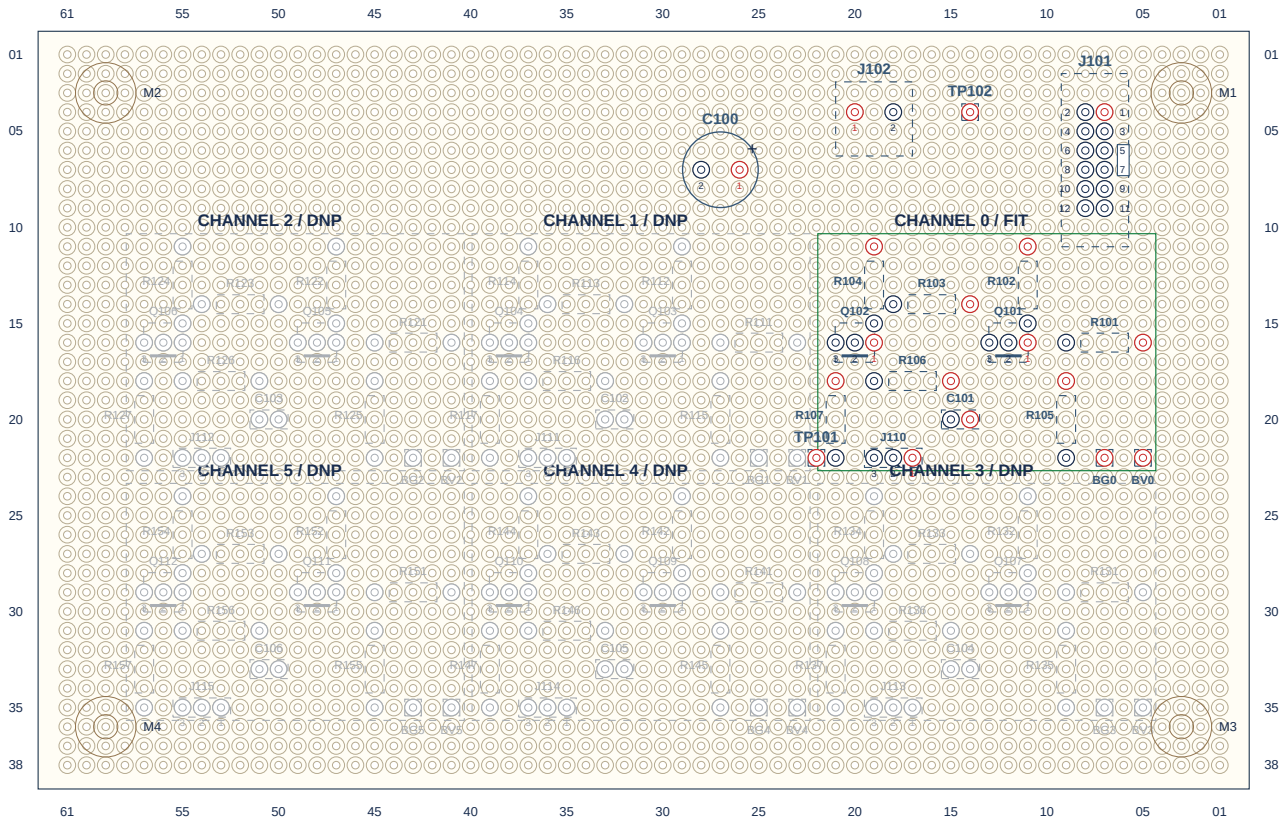
TO-92 leads are spread to a 2.54-mm row. Verify actual C/B/E order before soldering.

Print at 100% / Actual size. Disable Fit to page. Check the 50-mm bar and actual board-hole alignment.

50 mm scale check

Servo-output module: solder-side reference

Mirrored horizontally. Column61 is at left, column01 at right. Physical hole IDs do not change.



Wire crossings are insulated passovers, NOT electrical junctions. Only the listed endpoints connect.

SOLDER-SIDE ORIENTATION

Flip the board like turning a page.

Rows still run top to bottom.

IDC notch is on the right in this view.

Servo header: 1 signal, 2 +VS, 3 GND.

Verify cable polarity before insertion.

BVx/BGx are existing solder-pad nodes, not extra electronic components.

Use the listed heavy/load connections; no motor current through IDC grounds.

C100 positive = pad1, negative = pad2.

Grey bodies/pads remain unpopulated.

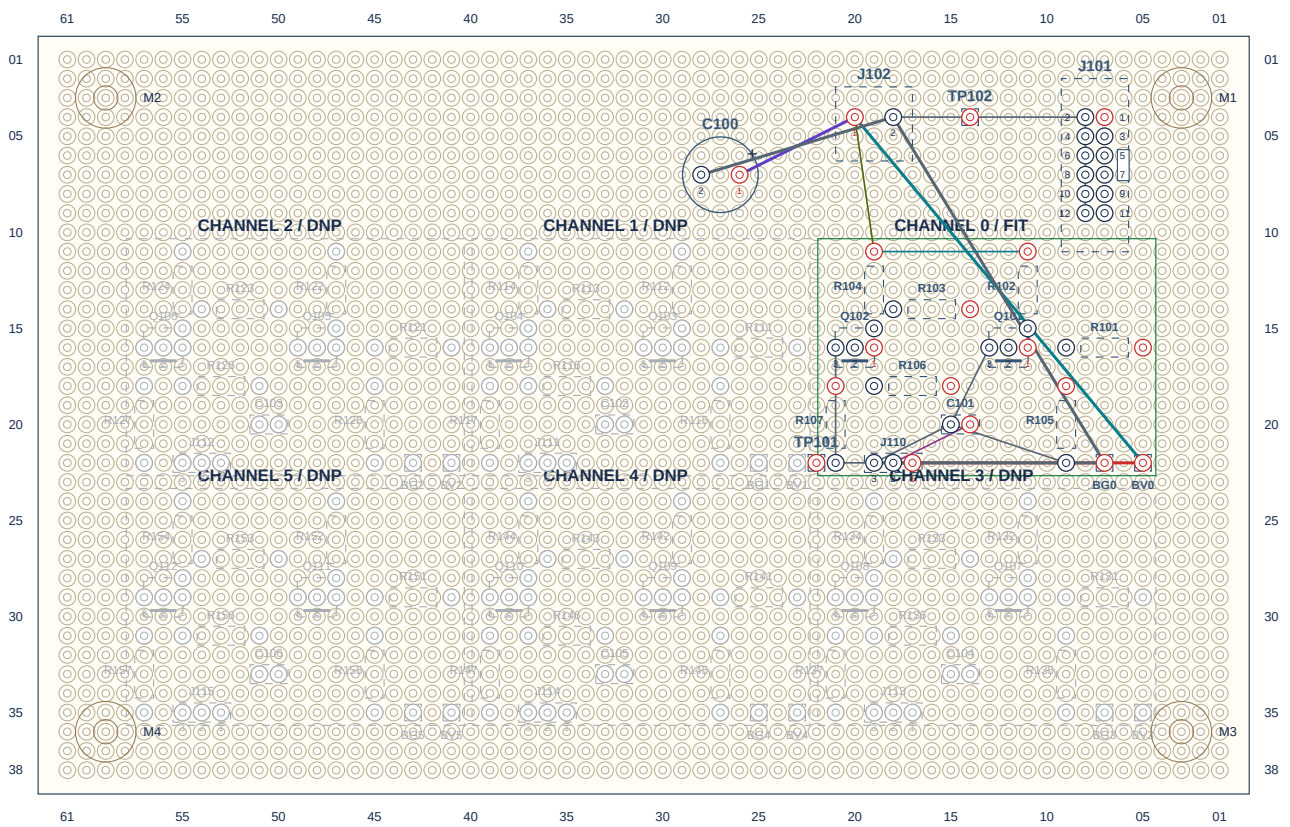
Print at 100% / Actual size. Disable Fit to page. Check the 50-mm bar and actual board-hole alignment.

50 mm scale check

Solder-side wiring: FIRST BUILD: power / ground

Mirrored view. Endpoint guide only: route insulated wires clear of sharp leads and mounting hardware.

P1 - UNBUILT / FIRST BUILD = ONE SERVO



WIRE	FROM / HOLE	TO / HOLE
W001*	J102.1 C20-R04	BV0.1 C05-R22
W002*	BV0.1 C05-R22	J110.2 C18-R22
W003*	J102.1 C20-R04	C100.1 C26-R07
W004*	J102.2 C18-R04	BG0.1 C07-R22
W005*	BG0.1 C07-R22	J110.3 C19-R22
W006*	J102.2 C18-R04	C100.2 C28-R07
W008	J110.2 C18-R22	C101.1 C14-R20
W009	J102.1 C20-R04	R104.1 C19-R11
W010	R104.1 C19-R11	R102.1 C11-R11
W011	J101.2 C08-R04	J101.4 C08-R05
W012	J101.4 C08-R05	J101.6 C08-R06
W013	J101.6 C08-R06	J101.8 C08-R07
W014	J101.8 C08-R07	J101.10 C08-R08
W015	J101.10 C08-R08	J101.12 C08-R09
W016	J101.2 C08-R04	TP102.1 C14-R04
W017	TP102.1 C14-R04	J102.2 C18-R04
W018	J110.3 C19-R22	R107.2 C21-R22
W019	J110.3 C19-R22	C101.2 C15-R20
W020	C101.2 C15-R20	Q101.3 C13-R16
W021	R107.2 C21-R22	Q102.3 C21-R16
W022	C101.2 C15-R20	R105.2 C09-R22

* = dedicated load-current wire; use a suitably rated reinforced conductor, not resistor leads.

FUTURE pages are space/wire reservations only. Do not power six servos from the present supply.

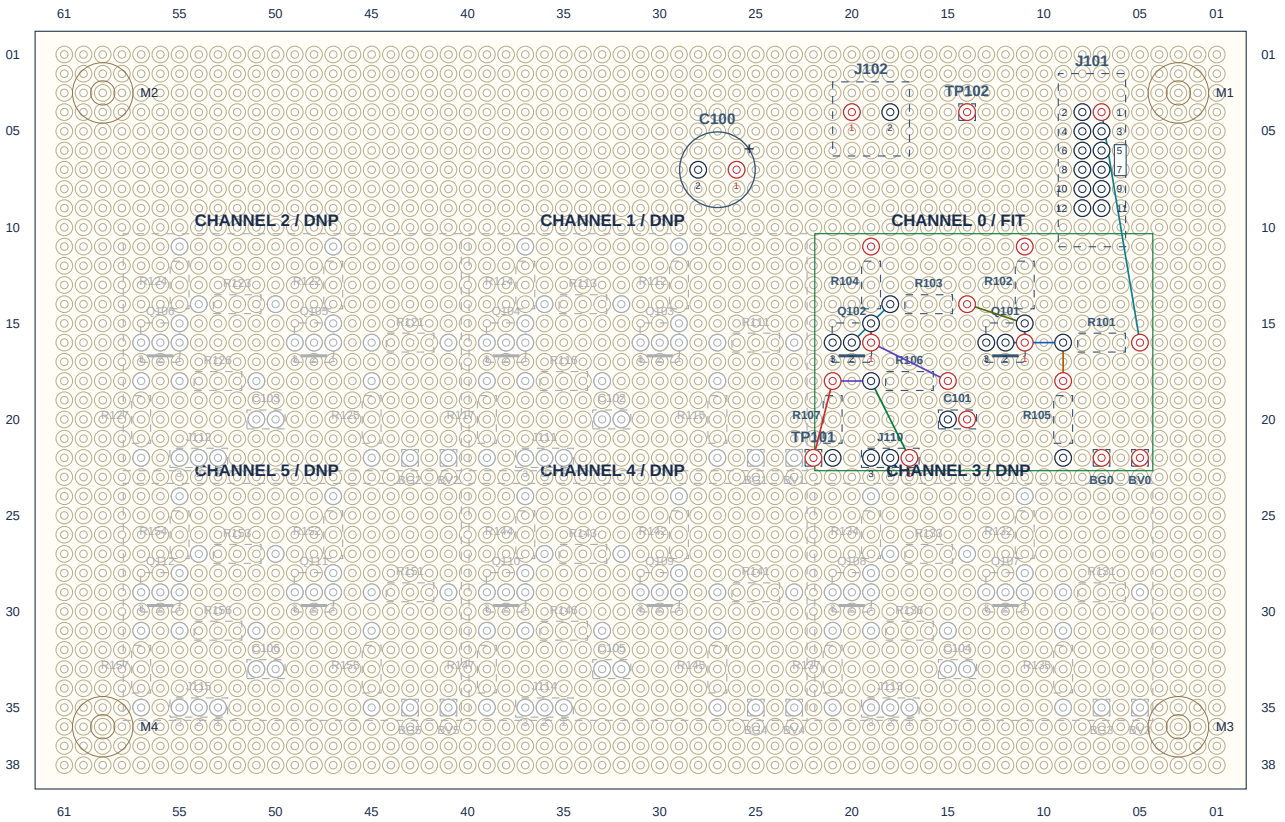
Print at 100% / Actual size. Disable Fit to page. Check the 50-mm bar and actual board-hole alignment.

50 mm scale check

Solder-side wiring: FIRST BUILD: channel0 logic

Mirrored view. Endpoint guide only: route insulated wires clear of sharp leads and mounting hardware.

P1 - UNBUILT / FIRST BUILD = ONE SERVO



WIRE	FROM / HOLE	TO / HOLE
W007	J101.1 C07-R04	R101.1 C05-R16
W023	TP101.1 C22-R22	R107.1 C21-R18
W024	R107.1 C21-R18	R106.2 C19-R18
W025	R106.2 C19-R18	J110.1 C17-R22
W026	R101.2 C09-R16	R105.1 C09-R18
W027	R101.2 C09-R16	Q101.2 C12-R16
W028	R102.2 C11-R15	Q101.1 C11-R16
W029	R102.2 C11-R15	R103.1 C14-R14
W030	R103.2 C18-R14	Q102.2 C20-R16
W031	R104.2 C19-R15	Q102.1 C19-R16
W032	Q102.1 C19-R16	R106.1 C15-R18

* = dedicated load-current wire; use a suitably rated reinforced conductor, not resistor leads.

FUTURE pages are space/wire reservations only. Do not power six servos from the present supply.

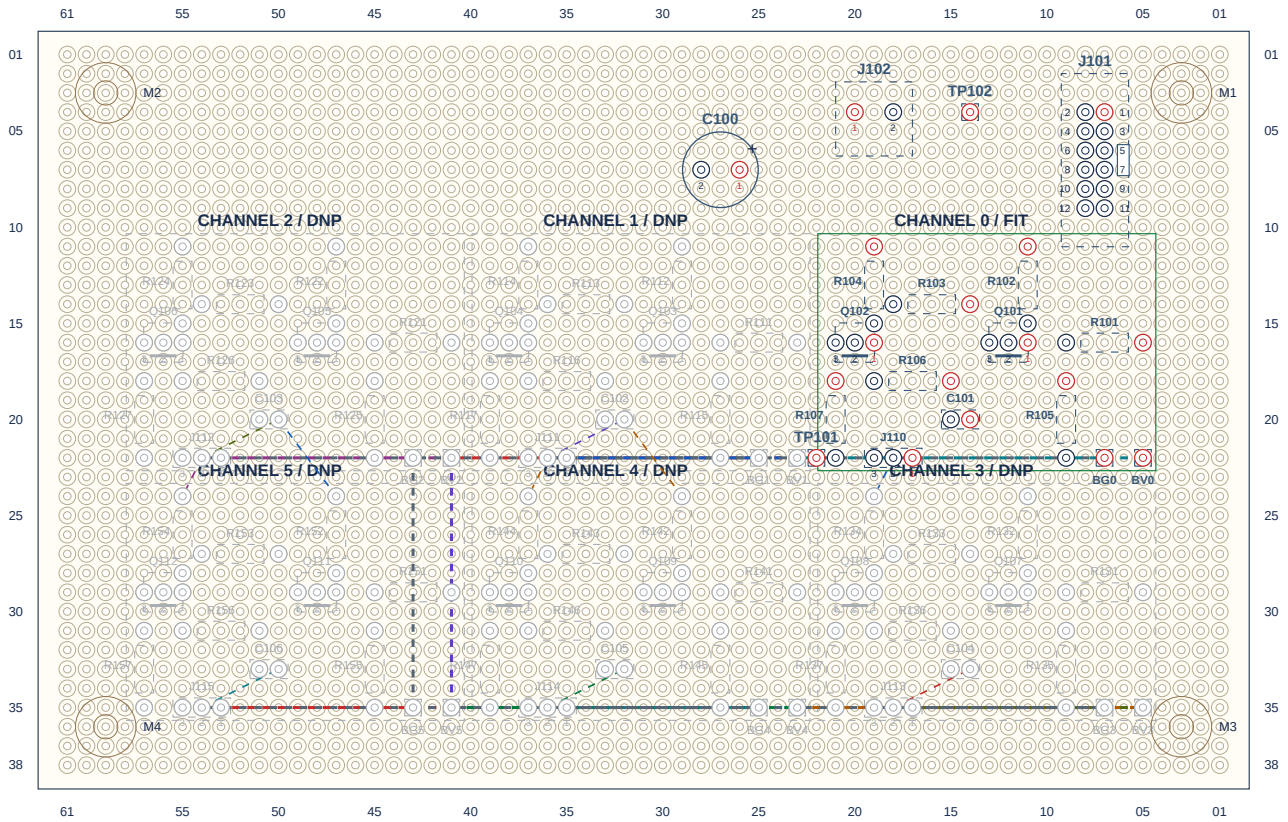
Print at 100% / Actual size. Disable Fit to page. Check the 50-mm bar and actual board-hole alignment.

50 mm scale check

Solder-side wiring: FUTURE ONLY: power expansion / 1

Mirrored view. Endpoint guide only: route insulated wires clear of sharp leads and mounting hardware.

P1 - UNBUILT / FIRST BUILD = ONE SERVO



WIRE	FROM / HOLE	TO / HOLE
W033*	BV0.1 C05-R22	BV1.1 C23-R22
W034*	BV1.1 C23-R22	BV2.1 C41-R22
W035*	BV2.1 C41-R22	BV5.1 C41-R35
W036*	BV5.1 C41-R35	BV4.1 C23-R35
W037*	BV4.1 C23-R35	BV3.1 C05-R35
W038*	BV1.1 C23-R22	J111.2 C36-R22
W039*	BV2.1 C41-R22	J112.2 C54-R22
W040*	BV3.1 C05-R35	J113.2 C18-R35
W041*	BV4.1 C23-R35	J114.2 C36-R35
W042*	BV5.1 C41-R35	J115.2 C54-R35
W043*	BG0.1 C07-R22	BG1.1 C25-R22
W044*	BG1.1 C25-R22	BG2.1 C43-R22
W045*	BG2.1 C43-R22	BG5.1 C43-R35
W046*	BG5.1 C43-R35	BG4.1 C25-R35
W047*	BG4.1 C25-R35	BG3.1 C07-R35
W048*	BG1.1 C25-R22	J111.3 C37-R22
W049*	BG2.1 C43-R22	J112.3 C55-R22
W050*	BG3.1 C07-R35	J113.3 C19-R35
W051*	BG4.1 C25-R35	J114.3 C37-R35
W052*	BG5.1 C43-R35	J115.3 C55-R35
W058	J111.2 C36-R22	R144.1 C37-R24
W059	J110.2 C18-R22	R134.1 C19-R24
W060	J112.2 C54-R22	R154.1 C55-R24
W061	J112.2 C54-R22	C103.1 C50-R20
W062	J115.2 C54-R35	C106.1 C50-R33
W063	J113.2 C18-R35	C104.1 C14-R33
W064	J111.2 C36-R22	C102.1 C32-R20
W065	J114.2 C36-R35	C105.1 C32-R33
W066	C102.1 C32-R20	R142.1 C29-R24
W067	C103.1 C50-R20	R152.1 C47-R24

* = dedicated load-current wire; use a suitably rated reinforced conductor, not resistor leads.

FUTURE pages are space/wire reservations only. Do not power six servos from the present supply.

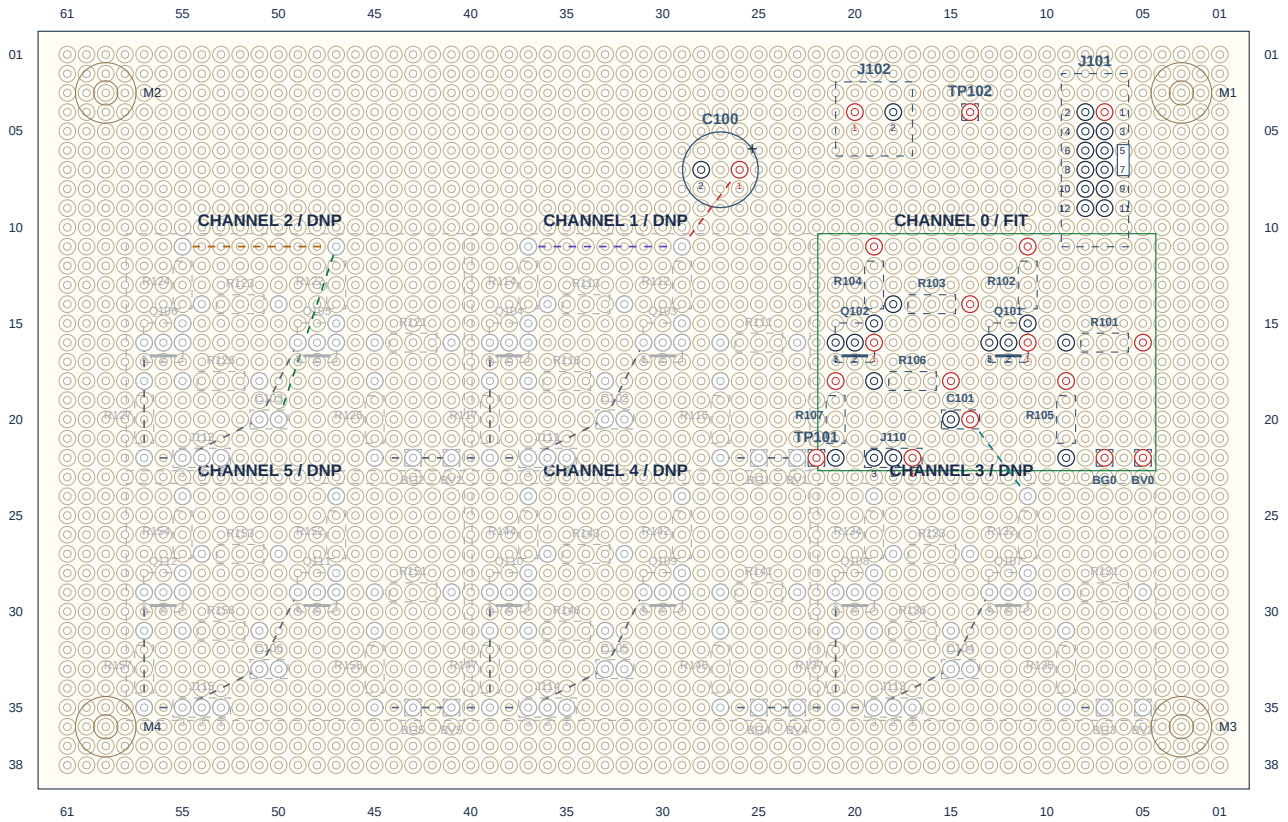
Print at 100% / Actual size. Disable Fit to page. Check the 50-mm bar and actual board-hole alignment.

50 mm scale check

Solder-side wiring: FUTURE ONLY: power expansion / 2

Mirrored view. Endpoint guide only: route insulated wires clear of sharp leads and mounting hardware.

P1 - UNBUILT / FIRST BUILD = ONE SERVO



WIRE	FROM / HOLE	TO / HOLE
W068	C101.1 C14-R20	R132.1 C11-R24
W069	C100.1 C26-R07	R112.1 C29-R11
W070	R112.1 C29-R11	R114.1 C37-R11
W071	C103.1 C50-R20	R122.1 C47-R11
W072	R122.1 C47-R11	R124.1 C55-R11
W073	J115.3 C55-R35	R157.2 C57-R35
W074	J112.3 C55-R22	R127.2 C57-R22
W075	J114.3 C37-R35	R147.2 C39-R35
W076	J113.3 C19-R35	R137.2 C21-R35
W077	J111.3 C37-R22	R117.2 C39-R22
W078	BG3.1 C07-R35	R135.2 C09-R35
W079	J111.3 C37-R22	C102.2 C33-R20
W080	C102.2 C33-R20	Q103.3 C31-R16
W081	J114.3 C37-R35	C105.2 C33-R33
W082	C105.2 C33-R33	Q109.3 C31-R29
W083	J113.3 C19-R35	C104.2 C15-R33
W084	C104.2 C15-R33	Q107.3 C13-R29
W085	J112.3 C55-R22	C103.2 C51-R20
W086	C103.2 C51-R20	Q105.3 C49-R16
W087	J115.3 C55-R35	C106.2 C51-R33
W088	C106.2 C51-R33	Q111.3 C49-R29
W089	R127.2 C57-R22	Q106.3 C57-R16
W090	R117.2 C39-R22	Q104.3 C39-R16
W091	R147.2 C39-R35	R155.2 C45-R35
W092	R147.2 C39-R35	Q110.3 C39-R29
W093	R107.2 C21-R22	R115.2 C27-R22
W094	R137.2 C21-R35	R145.2 C27-R35
W095	R137.2 C21-R35	Q108.3 C21-R29
W096	R157.2 C57-R35	Q112.3 C57-R29
W097	R117.2 C39-R22	R125.2 C45-R22

* = dedicated load-current wire; use a suitably rated reinforced conductor, not resistor leads.

FUTURE pages are space/wire reservations only. Do not power six servos from the present supply.

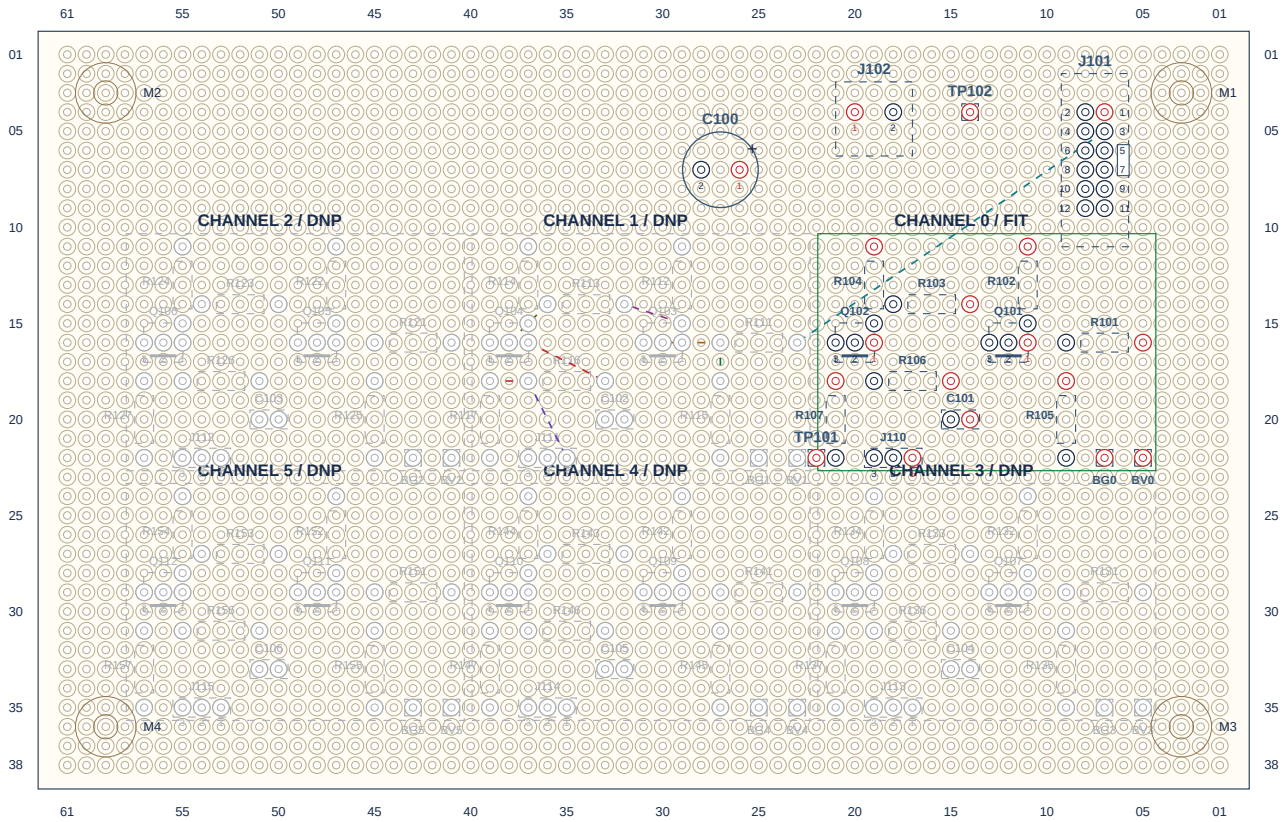
Print at 100% / Actual size. Disable Fit to page. Check the 50-mm bar and actual board-hole alignment.

50 mm scale check

Solder-side wiring: FUTURE ONLY: channel1 logic

Mirrored view. Endpoint guide only: route insulated wires clear of sharp leads and mounting hardware.

P1 - UNBUILT / FIRST BUILD = ONE SERVO



WIRE	FROM / HOLE	TO / HOLE
W053	J101.3 C07-R05	R111.1 C23-R16
W098	R116.2 C37-R18	R117.1 C39-R18
W099	R116.2 C37-R18	J111.1 C35-R22
W108	R111.2 C27-R16	R115.1 C27-R18
W109	R111.2 C27-R16	Q103.2 C30-R16
W110	R112.2 C29-R15	Q103.1 C29-R16
W111	R112.2 C29-R15	R113.1 C32-R14
W112	R113.2 C36-R14	Q104.2 C38-R16
W113	R114.2 C37-R15	Q104.1 C37-R16
W114	Q104.1 C37-R16	R116.1 C33-R18

* = dedicated load-current wire; use a suitably rated reinforced conductor, not resistor leads.

FUTURE pages are space/wire reservations only. Do not power six servos from the present supply.

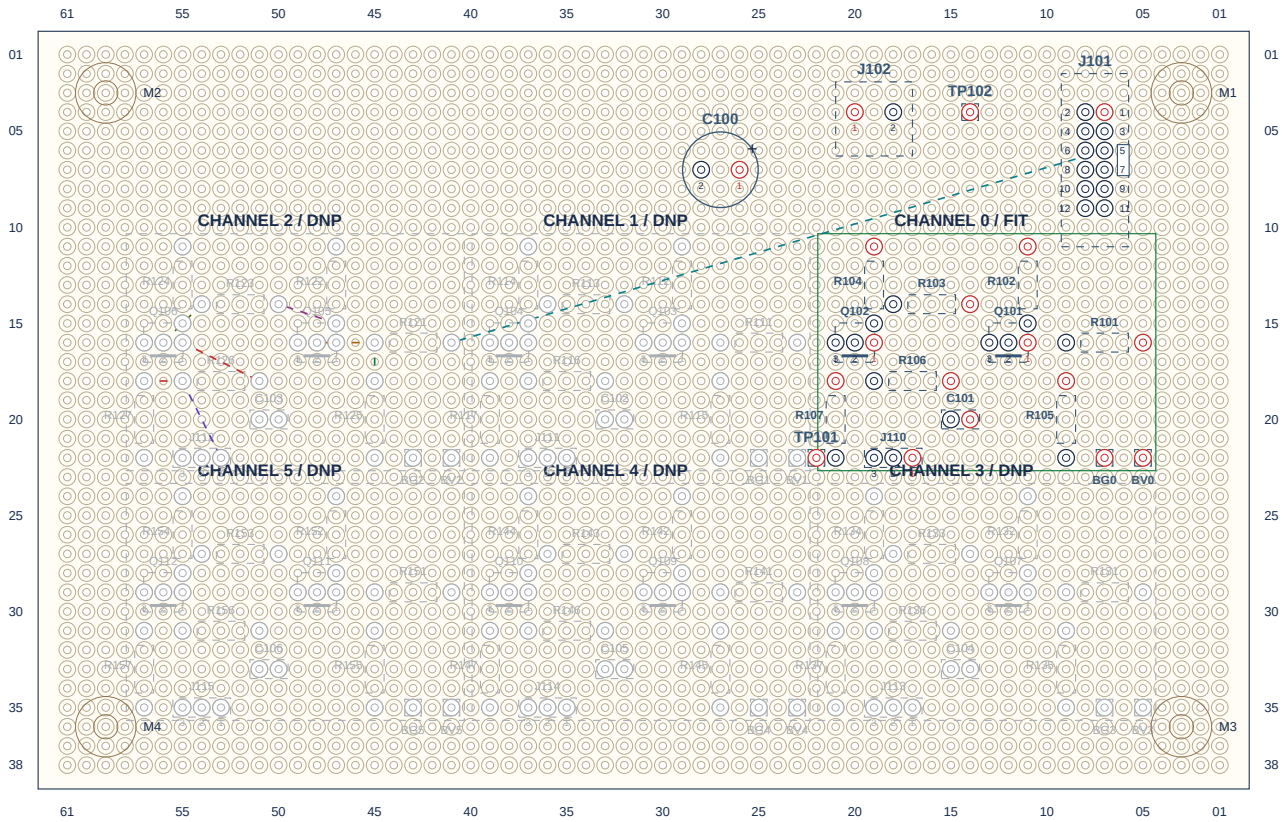
Print at 100% / Actual size. Disable Fit to page. Check the 50-mm bar and actual board-hole alignment.

50 mm scale check

Solder-side wiring: FUTURE ONLY: channel2 logic

Mirrored view. Endpoint guide only: route insulated wires clear of sharp leads and mounting hardware.

P1 - UNBUILT / FIRST BUILD = ONE SERVO



WIRE	FROM / HOLE	TO / HOLE
W054	J101.5 C07-R06	R121.1 C41-R16
W100	R126.2 C55-R18	R127.1 C57-R18
W101	R126.2 C55-R18	J112.1 C53-R22
W115	R121.2 C45-R16	R125.1 C45-R18
W116	R121.2 C45-R16	Q105.2 C48-R16
W117	R122.2 C47-R15	Q105.1 C47-R16
W118	R122.2 C47-R15	R123.1 C50-R14
W119	R123.2 C54-R14	Q106.2 C56-R16
W120	R124.2 C55-R15	Q106.1 C55-R16
W121	Q106.1 C55-R16	R126.1 C51-R18

* = dedicated load-current wire; use a suitably rated reinforced conductor, not resistor leads.

FUTURE pages are space/wire reservations only. Do not power six servos from the present supply.

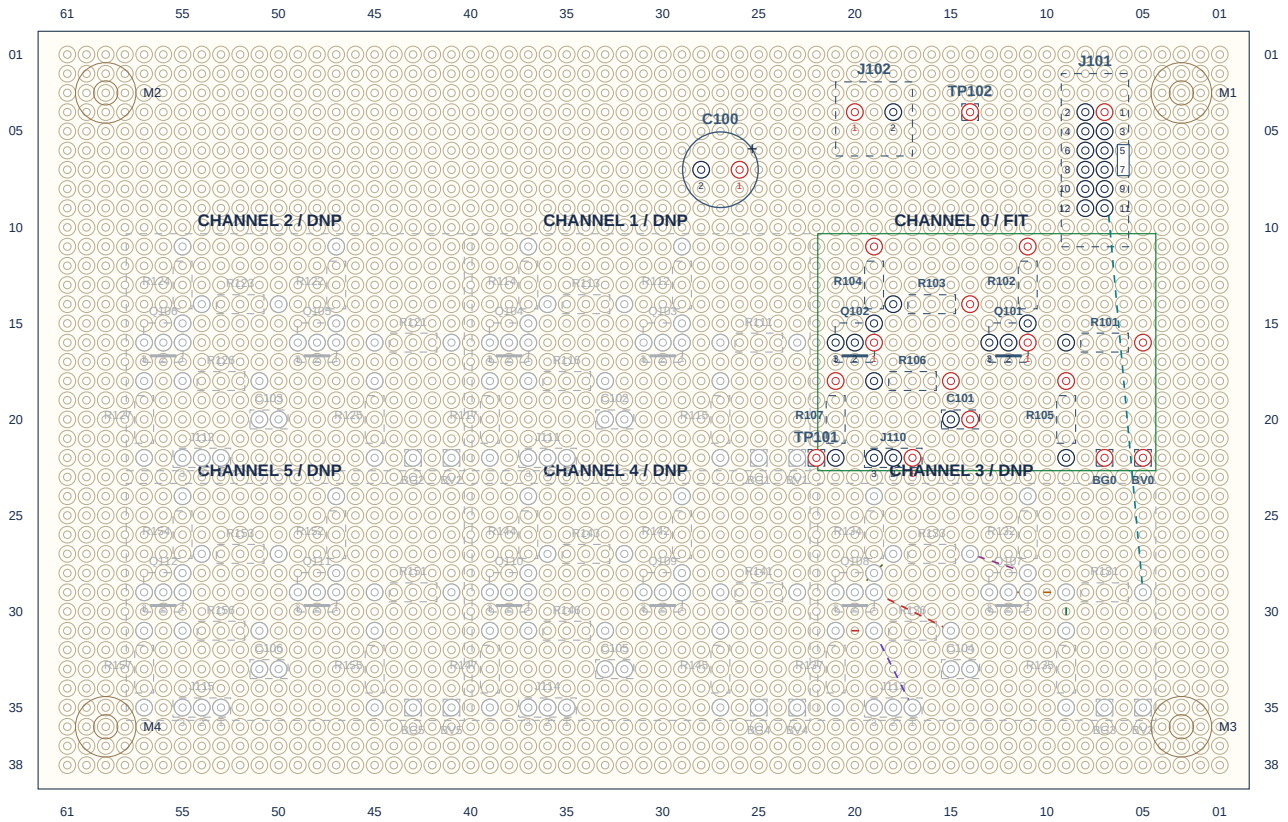
Print at 100% / Actual size. Disable Fit to page. Check the 50-mm bar and actual board-hole alignment.

50 mm scale check

Solder-side wiring: FUTURE ONLY: channel3 logic

Mirrored view. Endpoint guide only: route insulated wires clear of sharp leads and mounting hardware.

P1 - UNBUILT / FIRST BUILD = ONE SERVO



WIRE	FROM / HOLE	TO / HOLE
W055	J101.7 C07-R07	R131.1 C05-R29
W102	R136.2 C19-R31	R137.1 C21-R31
W103	R136.2 C19-R31	J113.1 C17-R35
W122	R131.2 C09-R29	R135.1 C09-R31
W123	R131.2 C09-R29	Q107.2 C12-R29
W124	R132.2 C11-R28	Q107.1 C11-R29
W125	R132.2 C11-R28	R133.1 C14-R27
W126	R133.2 C18-R27	Q108.2 C20-R29
W127	R134.2 C19-R28	Q108.1 C19-R29
W128	Q108.1 C19-R29	R136.1 C15-R31

* = dedicated load-current wire; use a suitably rated reinforced conductor, not resistor leads.

FUTURE pages are space/wire reservations only. Do not power six servos from the present supply.

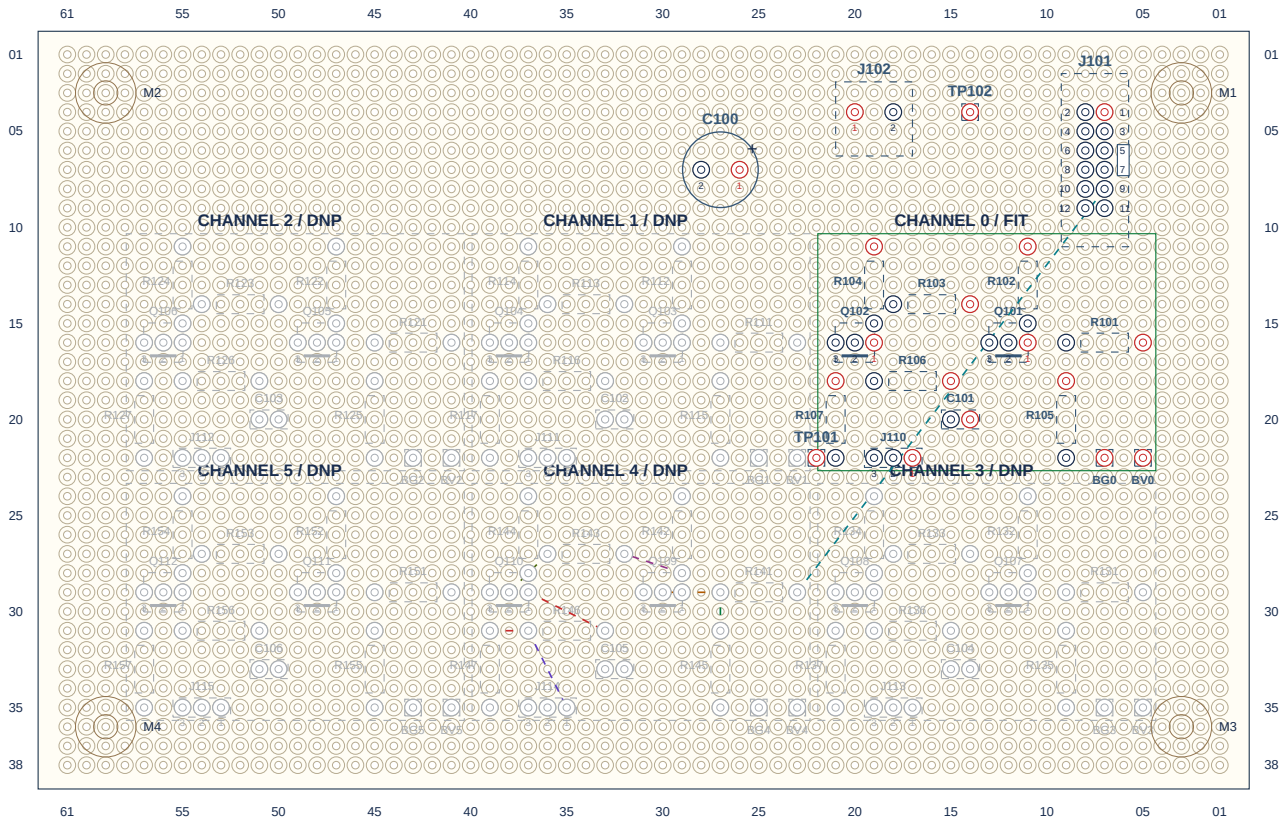
Print at 100% / Actual size. Disable Fit to page. Check the 50-mm bar and actual board-hole alignment.

50 mm scale check

Solder-side wiring: FUTURE ONLY: channel4 logic

Mirrored view. Endpoint guide only: route insulated wires clear of sharp leads and mounting hardware.

P1 - UNBUILT / FIRST BUILD = ONE SERVO



WIRE	FROM / HOLE	TO / HOLE
W056	J101.9 C07-R08	R141.1 C23-R29
W104	R146.2 C37-R31	R147.1 C39-R31
W105	R146.2 C37-R31	J114.1 C35-R35
W129	R141.2 C27-R29	R145.1 C27-R31
W130	R141.2 C27-R29	Q109.2 C30-R29
W131	R142.2 C29-R28	Q109.1 C29-R29
W132	R142.2 C29-R28	R143.1 C32-R27
W133	R143.2 C36-R27	Q110.2 C38-R29
W134	R144.2 C37-R28	Q110.1 C37-R29
W135	Q110.1 C37-R29	R146.1 C33-R31

* = dedicated load-current wire; use a suitably rated reinforced conductor, not resistor leads.

FUTURE pages are space/wire reservations only. Do not power six servos from the present supply.

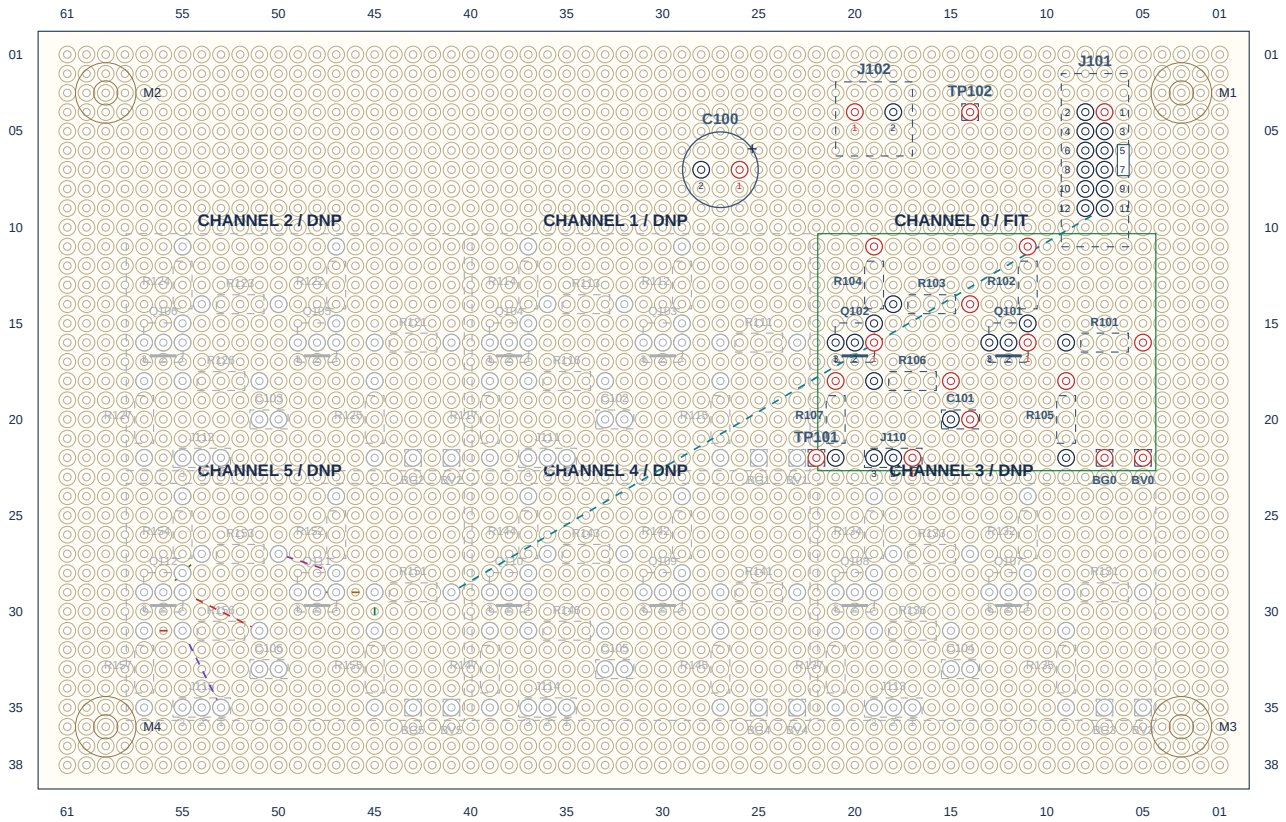
Print at 100% / Actual size. Disable Fit to page. Check the 50-mm bar and actual board-hole alignment.

50 mm scale check

Solder-side wiring: FUTURE ONLY: channel5 logic

Mirrored view. Endpoint guide only: route insulated wires clear of sharp leads and mounting hardware.

P1 - UNBUILT / FIRST BUILD = ONE SERVO



WIRE	FROM / HOLE	TO / HOLE
W057	J101.11 C07-R09	R151.1 C41-R29
W106	R156.2 C55-R31	R157.1 C57-R31
W107	R156.2 C55-R31	J115.1 C53-R35
W136	R151.2 C45-R29	R155.1 C45-R31
W137	R151.2 C45-R29	Q111.2 C48-R29
W138	R152.2 C47-R28	Q111.1 C47-R29
W139	R152.2 C47-R28	R153.1 C50-R27
W140	R153.2 C54-R27	Q112.2 C56-R29
W141	R154.2 C55-R28	Q112.1 C55-R29
W142	Q112.1 C55-R29	R156.1 C51-R31

* = dedicated load-current wire; use a suitably rated reinforced conductor, not resistor leads.

FUTURE pages are space/wire reservations only. Do not power six servos from the present supply.

Print at 100% / Actual size. Disable Fit to page. Check the 50-mm bar and actual board-hole alignment.

50 mm scale check